QUESTIONS 10

1) Consider the following system.

$$\begin{aligned} 4x_1 + x_2 - x_3 + x_4 &= -2 \\ x_1 + 4x_2 - x_3 - x_4 &= -1 \\ -x_1 - x_2 + 5x_3 + x_4 &= 0 \\ x_1 - x_2 + x_3 + 3x_4 &= 1 \end{aligned}$$

a) Find $x^{(2)}$ using the Jacobi method with $x_1^{(0)} = 0$, $x_2^{(0)} = 0$ and $x_3^{(0)} = 0$.
b) Find Jacobi matrix for this problem.

2) Consider the following system.

$$\begin{aligned} x_1 + 2x_2 - 2x_3 &= 7 \\ x_1 + x_2 + x_3 &= 2 \\ 2x_1 + 2x_2 + x_3 &= 5 \end{aligned}$$

a) Find $x^{(2)}$ using the Gauss-Seidel iterative method with $x_1^{(0)} = 0$, $x_2^{(0)} = 0$ and $x_3^{(0)} = 0$.
b) Determine whether Jacobi and Gauss-Seidel methods are convergent for this system.